IE 7200: Supply Chain Engineering — Individual Term Project

Monday Dispatch: Applied Logistics Optimization for Northeastern Home Goods

Northeastern University | Department of Mechanical & Industrial Engineering
Online / Asynchronous Section

1. Executive Summary and Project Philosophy

This term you are the Forward Deployed Engineer assigned to Modern Analytics Dynamics, a consulting practice that builds algorithmic decision engines for complex supply chains. Your client is Northeastern Home Goods, a regional furniture retailer distributing from a single centralised distribution centre in Ayer, Massachusetts. As sales volume has climbed, NHG's manual dispatching rules have reached breaking point: dispatchers are buying expensive spot capacity, drivers are brushing against Department of Transportation duty ceilings, and average vehicle volume utilisation sits below sixty percent.

You own **Monday dispatch**. Your instance contains **43 store delivery orders across 34 distinct store locations**, totalling **10,223 cubic feet** of freight. Every other weekday in the week is owned by a different engineer, and the week decomposes cleanly: because each order carries a fixed required delivery day and no constraint couples one day to another, Monday can be solved in complete isolation and the weekly total is simply the sum of the five daily plans. Your result is one fifth of a real answer, not a toy.

Be clear-eyed about what follows. A multi-stop capacitated vehicle routing problem with 34 customer locations, binding DOT duty limits, a ten-hour delivery window and a constrained loading dock is not going to be solved to proven mathematical optimality this term. That is not a defect in the assignment; it is how industrial logistics behaves. What you are graded on is the rigour of your engineering, the honesty of your reporting, and your ability to say how far from optimal your plan actually is.

2. Your Role and the Client Engagement Model

Treat the term as an active consulting engagement conducted remotely. You are the entire analytics pod: there is no teammate to hand the modelling to while you build slides, and no teammate whose code you can take on faith. The MAD VP of Logistics Operations, played by your instructor, is your technical client and will probe runtime scaling, bound validity and DOT compliance. The NHG VP of Supply Chain, played by the teaching assistant, is your executive stakeholder and cares about landed cost, route reliability and driver retention.

Because this section is asynchronous and you are working full time, every client touchpoint is asynchronous by default. You submit a short recorded walkthrough and a written artefact; your client responds in writing within three business days. Live office hours are available on request but are never required. You are expected to work independently, and the cohort's five datasets are deliberately different so that comparing final numbers with a classmate is meaningless. Discussing methods, libraries and debugging strategy with peers is encouraged and should be acknowledged; sharing code or results is not.

3. Milestone Cadence and Roadmap

The schedule is anchored on four submissions plus a final package. Each phase extends the previous one, and nothing you build is discarded.

Milestone	Week	Format	Core Technical Objective	Primary Artefact
M0: Scoping	4	Written memo, instructor responds in writing	Fix problem boundaries, state your algorithmic roadmap and your definition of a feasible route	1-page Scope Memo
M1: Baseline Optimiser	9	4-minute recorded code walkthrough plus artefact	Working CVRP solver for the homogeneous Van fleet with time window and DOT rules, feasibility-validated	Code v0.1 plus benchmark table
M2: Fleet and Dock Scheduling	12	4-minute recorded demo plus memo	Mixed Van and Straight Truck fleet, plus dock-wave departure scheduling	Code v0.2 plus 2-page Decision Memo
M3: Continuous Improvement (optional bonus)	13	Written insight memo	Lower bound, optimality gap and a convergence chart	Code v0.3 plus 2-page Insight Memo
Final	14-15	8-minute recorded pitch	Production-grade package and executive narrative	Repo, report and recorded pitch

4. Technical Scope and Architecture

4.1 Show the journey, not only the destination

It is not acceptable to present only a final number. Routing engineers do not arrive at a production schedule on the first attempt; they get there through a sequence of prototypes, each exposing a bottleneck the previous one hid. Your memos must document that progression: what you tried, what each method cost in runtime and complexity, what it bought in mileage, and why you moved on. A student who reports a construction heuristic at 2,480 miles, local search at 2,190 miles and a tuned metaheuristic at 2,095 miles tells a credible engineering story (those figures are illustrative only and are not targets for any day). A student who reports only the last figure is asking the client for blind faith.

The methodological ladder runs from brute-force enumeration on a deliberately tiny subset of six to nine stops, which is the only place you will ever know the true global optimum and can measure heuristic error exactly; through a construction heuristic such as Clarke-Wright savings, sweep or nearest-neighbour adapted to capacity and duty rules; through

local search using intra-route two-opt and or-opt moves together with inter-route relocate and swap, which typically returns the highest improvement per line of code written; and finally to a metaheuristic such as simulated annealing, tabu search or large neighbourhood search layered over that local search. There is no mandated path. The requirement is comparative: benchmark at least two distinct methods against each other by M1, and at least two more, or a substantial enhancement of an existing one, by M2.

4.2 Milestone 1: the baseline optimiser

Build an end-to-end solver for NHG's homogeneous Van fleet, where each van carries 3,200 cubic feet, unloads at 0.03 minutes per cubic foot subject to a 30-minute minimum per stop, and travels at 40 miles per hour. A route is feasible only if it satisfies all four of the following simultaneously. Vehicle volume must not exceed 3,200 cubic feet. Driving time, computed as 1.5 minutes per mile over the whole tour including both depot legs, must not exceed 660 minutes. On-duty time, defined as driving plus unload, must not exceed 840 minutes. Finally, the in-window span, defined as inter-stop driving plus every unload on the route, must fit inside the 600-minute delivery window from 8:00 AM to 6:00 PM, because the first delivery may not begin before 8:00 AM and the last must be complete by 6:00 PM, while the drive out to the first stop and back from the last happens outside that window.

That fourth constraint deserves care. It is a property of the route's *sequence*, not merely of the set of stops, and it is not minimised by the cheapest tour. Minimising total miles includes the two depot legs, so the cheapest ordering can carry more inter-stop mileage — and therefore a longer span — than a slightly more expensive ordering. A set of stops is feasible if **some** ordering satisfies the limits, so testing only your cost-optimal tour will cause you to reject routes that are in fact legal.

Design your solver as a parameterised, importable module from the first commit, never as a script with constants edited by hand at the top. In M2 you will call it across fleet configurations, and if you attempt M3 you will invoke it thousands of times inside benchmark loops.

4.3 Milestone 2: mixed fleet and dock-wave scheduling

M2 adds two dimensions of operational realism.

The first is **heterogeneous fleet assignment**. Some consignments are white-glove or oversized items that can only be handled from the smaller Straight Truck, which carries 1,400 cubic feet and unloads at 0.043 minutes per cubic foot; every other consignment rides either vehicle. On Monday 10 of your 43 orders carry that restriction, 1,576 cubic feet in total, so at least 2 straight trucks must be in your mix. Your solver must jointly assign vehicle types and report the recommended fleet mix.

Read the restriction carefully, because it is a property of the **order, not of the location**. A store can receive one restricted order and one unrestricted order on the same day, and on your instance 3 locations do. If you lift the flag up to the location and force every order there onto a straight truck, you will build a model that is more constrained than the operation actually is. The flag ships as `OrderRequiresStraightTruck` in your instance file, and as the `RequiresStraightTruck` column of the vehicle-type table described in Section 5.

The second is **dock-wave departure scheduling**, and it is specific to this online section. The Ayer depot cannot release its whole fleet at once. It operates `DockBays = 2` loading bays; the first dispatch wave departs at `DockOpenMin = 05:00`; and each bay needs `DockLoadMinPerVehicle = 90` minutes to load and turn around. Vehicles therefore depart in waves of two at 05:00, 06:30, 08:00 and so on. A vehicle dispatched in a later wave reaches its first stop later, and since the last delivery must still finish by 6:00 PM, its usable in-window span shrinks accordingly. Formally, if a vehicle departs at time `dep` and takes `travel` minutes to reach its first stop, service there begins no earlier than `max(480, dep + travel)`, and the route's span must fit between that moment and minute 1080.

This couples fleet size to feasibility in a way the M1 model does not capture: adding a route is no longer merely more miles, it pushes some vehicle into a later wave and may make the plan infeasible. Your decision memo must state how many routes your plan uses, which wave each takes, and how much slack remains on the tightest one.

4.4 Milestone 3: continuous improvement (optional bonus)

Everything here is optional, and delivering M1, M2 and the final pitch with rigour earns an A. M3 exists for students who want to know how good their plan actually is, and it is the part of this project that shows up most vividly in technical interviews.

You may pursue a batch scenario engine that runs multiple operational scenarios without manual edits; a provable lower bound on your day's instance with a verified optimality gap; or a set-partitioning formulation solved by column generation, whose LP relaxation gives the strongest bound available and whose honest post-mortem on where the pricing subproblem struggled is the most marketable artefact in this course. Depth in one direction earns more credit than a shallow pass at all three. Whichever you choose, one visual should anchor the work: a convergence plot with runtime on the horizontal axis, your best feasible solution descending, your lower bound ascending, and the optimality gap visibly narrowing.

The standard for insight is concrete. "Our plan looks good" is an opinion. "Our plan sits within 6.4 percent of a proven lower bound, and most of that spread comes from two long-haul routes that no amount of local search can merge" is an engineering finding.

5. Your Dataset

5.1 The instance at a glance

Quantity	Value
Delivery orders	43
Distinct store locations	34
Total volume	10,223 ft³
Total unload time	1,324.7 min
Minimum vans by volume alone	4
Locations receiving more than one order	8
Farthest store from the depot	201 miles
Vehicle capacity	3,200 ft³ (Van), 1,400 ft³ (Straight Truck)
Dock	2 bays, first wave 05:00, 90 min per vehicle

5.2 Two equivalent file formats

Production planning data rarely arrives in the shape a solver wants, and learning to ingest it on its own terms is part of this assignment. Your instance is therefore published twice, in the two formats you will actually meet in practice. They carry identical content: the CSV tables are generated from the JSON and have been verified to reconstruct every field of it exactly. Read whichever suits your stack and ignore the other.

Path	Shape	Contents
`data/ie7200_Mon.json`	one object	Every parameter, order array and the distance matrix in a single file
`data/csv/ie7200_Mon_parameters.csv`	18 rows	One row per scalar parameter, with its units and meaning
`data/csv/ie7200_Mon_orders.csv`	43 rows	One row per delivery order
`data/csv/ie7200_Mon_vehicle_types.csv`	43 rows	Which orders need a Straight Truck, for Milestone 2
`data/csv/ie7200_Mon_locations.csv`	35 rows	One row per location, the depot included
`data/csv/ie7200_Mon_distance.csv`	35 x 36	The distance matrix, its first column the origin index

The JSON is the canonical artefact and the CSV set is generated from it, so if the two ever disagree the JSON wins and you should report the discrepancy. The CSVs are plain comma-separated text with a single header row and no quoting, readable with `pandas.read_csv`, R's `read.csv`, or the `csv` module in the standard library.

5.3 Column reference

`orders.csv` carries one row per order, which is the unit you must cover exactly once.

Column	Meaning
`OrderIndex`	Row number, 0 to 42. Your local order identifier
`GlobalOrderId`	The same order's identifier in the master weekly dataset
`LocationIndex`	Row and column of this order's location in the distance matrix
`GlobalLocationId`	The same location's identifier in the master weekly dataset
`Volume_ft3`	Freight volume, charged against vehicle capacity
`UnloadTime_min`	Service time at the stop, already floored at 30 minutes

`vehicle_types.csv` carries the Milestone 2 fleet restriction, again one row per order and in the same row order as `orders.csv`. You can ignore it until M2.

Column	Meaning
`OrderIndex`	Matches `OrderIndex` in `orders.csv`
`GlobalOrderId`	The same order in the master weekly dataset
`LocationIndex`	Where the order is delivered
`GlobalLocationId`	The same location in the master weekly dataset
`Volume_ft3`	Repeated from `orders.csv` so this table stands alone
`RequiresStraightTruck`	1 if only a Straight Truck may carry it, 0 if either vehicle may
`AllowedVehicles`	The same fact in words

`locations.csv` carries one row per location. Its last three columns are derived from the other two tables and are shipped only for convenience; recomputing them is a sound first test of your ingestion code.

Column	Meaning
`LocationIndex`	Index into the distance matrix, 0 to 34
`GlobalLocationId`	The same location in the master weekly dataset
`IsDepot`	1 for the Ayer depot, 0 for a store
`DistanceFromDepot_mi`	Derived. Row 0 of the distance matrix
`OrderCount`	Derived. Orders due at this location
`TotalVolume_ft3`	Derived. Their combined volume

Index 0 in your distance matrix is the Ayer depot. Note carefully that an order and a stop are not the same thing: 43 orders arrive at 34 distinct store locations, so a vehicle that visits one location may discharge several orders there and pays the unload time of each. The `GlobalOrderId` and `GlobalLocationId` columns map your compacted indices back to the master weekly dataset, which you will need if you ever want to compare notes with the engineer who owns an adjacent day.

5.4 Two warnings about the data

Both of these reflect how real order books behave. First, the distance matrix is **symmetric but not metric**: there are triples of locations where travelling directly between two of them

is longer than routing through the third. Any reasoning that assumes the triangle inequality — including several textbook pruning rules — is unsafe here. Second, Monday presents the longest reach in the network — a store 201 miles from Ayer, far enough that a single round trip consumes most of an 11-hour driving day. Validate your structural assumptions against the data before you trust them; at least one property of your instance will break a model written from the textbook definition alone, and finding it is part of the assignment.

6. Working with AI Tools and Engineering Standards

AI assistance is permitted and encouraged, and must be disclosed. Maintain an `AI_USAGE.md` log recording which tools you used, for what, and what you changed afterwards. The standard is simple: you must be able to explain and defend every line you submit, including lines a model wrote for you. Cold-call questions on your own code are a normal part of evaluation.

Your repository should have a clear structure separating data ingestion, the solver module, experiments and tests; a `README.md` with a one-line reproduction command; a `runbook.md` covering environment setup, the standard weekly workflow and known limitations; and an automated feasibility test suite. A `pytest` suite that re-validates every route in your final plan against all four constraints is worth more to your grade than any amount of prose asserting that your plan is legal.

7. AI Disclosure

I used Claude to draft and edit portions of this document. I set every policy, grading weight, and technical requirement here, reviewed every line, and am responsible for the result.

8. Grading

Component	Weight	Assessment Focus
M0: Scope memo	5%	Clarity of scope, a correct statement of route feasibility, concrete milestones
M1: Baseline optimiser	30%	Mathematical correctness, validated feasibility, clean modular code, a documented improvement path
M2: Fleet and dock scheduling	30%	Mixed-fleet logic, dock-wave scheduling, measured improvement over M1, quality of the decision memo
Final package and recorded pitch	25%	Narrative clarity, operational credibility, defensibility under written follow-up
Engineering practice and reproducibility	10%	Repository hygiene, one-command reproduction, test coverage, AI disclosure
Core total	**100%**	

Milestone 3 sits outside this scale and is worth up to ten points added to your final course

score at the instructor's discretion. Skipping it carries no penalty.

A submission whose reported numbers cannot be reproduced from its own repository, or whose plan fails feasibility validation, cannot receive full credit on that milestone regardless of how good the mileage figure looks. Reporting an honest, larger number beats reporting an attractive, infeasible one, and this will be enforced.

9. Final Deliverables

Your final package contains three artefacts. The **code repository** holds fully commented source, an automated ingestion path from the provided JSON, a one-line reproduction command, your test suite, `runbook.md` and `AI_USAGE.md`. The **final report**, eight to ten pages, analyses what your M1 to M2 extensions cost and saved, the comparative gain at each rung of your algorithmic ladder, where your approach is weakest, and what you would do with another two weeks; students who attempt M3 add their verified optimality gap and where the search plateaued. The **recorded pitch**, eight minutes with clear narration, ingests the raw order file and produces a full Monday schedule on screen, shows the mixed-fleet and dock-wave results with their tradeoffs visible, and reports utilisation and duty metrics.

10. Academic Integrity

The five datasets in this section are different by construction, so a matching number between two students is evidence rather than coincidence. Reusing code, formulations or benchmark figures from classmates, past course iterations, or online solutions without explicit documented attribution is a severe violation, and this applies equally whether or not an AI tool mediated the transfer. Disclosed AI use following Section 6 is fully permitted.

11. References

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